

LME Metals Risk Premia — Strategy Reference

LME Copper (2006–2025) | LME Aluminium (2006–2026) | July 2026

Common conventions: PnL = position × $\Delta F1_{\text{continuous}}$ (ratio back-adjusted). TC = 5 bps round-trip charged on $F1_{\text{raw}}$ at each flip. Same-Day: position_{t} = signal_{t-1} (shift-1, no look-ahead). Lag-1: position_{t} = signal_{t-2} (signal observed, then executed next day). Sharpe = annualised, active-day convention. All figures below re-verified against a live run of the dashboard engine (app.py) on 2026-07-05 — not carried over from any prior document.

Family / Variant	Signal Formula	Position Rule	Timing	OOS Category
TAB 7 — MOMENTUM	TAB 7 — MOMENTUM	TAB 7 — MOMENTUM	TAB 7 — MOMENTUM	TAB 7 — MOMENTUM
MA Crossover (grid search)	signal(t) = sign[MA($F1_{\text{raw}}$, m) – MA($F1_{\text{raw}}$, n)] Deployed: Cu MA(20,45), Al MA(60,115)	+1 long / –1 short	Same-Day ★ (Lag-1 also tested)	Fixed-param WF
MA Crossover — legacy grid optimum (Cu comparison only, not deployed)	Same formula, m=35, n=43 (8-day gap, ratio 1.23 — grid-search artifact, not a genuine multi-scale gap)	+1 / –1	Same-Day	Fixed-param WF
Anchors EW MA(10,25) + MA(35,43) + MA(63,100) (structural, not deployed)	pos_EW(t) = mean[sign(MA_fast,i – MA_slow,i)] for i=1,2,3	{–1, –2/3, –1/3, 0, +1/3, +2/3, +1}	Lag-1	Parameter-free
Anchors + IS-Opt Weights (same 3 anchors; Copper research line)	w* = argmax Sharpe(w; prior 5yr IS), w≥0, Σw=1 port(t) = Σ w*_i · sign(MA_fast,i – MA_slow,i), refit each window	Continuous (–1 to +1)	Lag-1	Rolling WF (true OOS)
CTA Single-scale (S, L) pairs — grid	z(t) = [$F1(t) - F1(t-L)$] / $\sigma(F1, S)$ pos(t) = sign(z(t))	+1 / –1 (no flat)	Same-Day	Fixed-param WF
CTA Baz-Granger Paper 3 speed tiers (published S,L)	z_i = [$F1(t) - F1(t-L_i)$] / $\sigma(F1, S_i)$ u_i = z_i · exp(–z_i ² /4) / 0.89 pos(t) = sign(mean(u_fast, u_med, u_slow))	Continuous; rescaled to unit variance	Same-Day (Cu) / Lag-1 (Al) — varies	Parameter-free
TAB 8 — CARRY	TAB 8 — CARRY	TAB 8 — CARRY	TAB 8 — CARRY	TAB 8 — CARRY
V1 Roll Yield — level ($F1-F2$)/ $F1$, ($F1-F3$)/ $F1$, (Cash–3M)/Cash	signal(t) = (Fa – Fb) / Fa	pos(t) = sign(signal(t)) +1 backwardation / –1 contango	Same-Day ★ (Lag-1 tested)	Parameter-free
V2 Long Slope — 10 tenor pairs F3–F15, F4–F16, ..., F12–F24	signal(t) = (Fj(t) – Fk(t)) / Fk(t)	pos(t) = sign(signal(t)) +1 / –1	Same-Day (Lag-1 tested)	Fixed-param WF
V3 Z-Score of Roll Yield (252d) [Deployed: Al tab default; both metals' Portfolio leg]	ry(t) = ($F1-F2$)/ $F1$ signal(t) = [ry(t) – $\mu_{252}(\text{ry})$] / $\sigma_{252}(\text{ry})$	pos(t) = sign(signal(t)) +1 / –1	Same-Day ★	Parameter-free
V4 Carry-Momentum — Δ roll yield N = 5, 10, 20, 60d [Deployed: Cu tab default]	ry(t) = ($F1-F2$)/ $F1$ signal(t) = ry(t) – ry(t–N)	pos(t) = sign(signal(t)) +1 / –1	Same-Day ★	Fixed-param WF
TAB 9 — VALUE	TAB 9 — VALUE	TAB 9 — VALUE	TAB 9 — VALUE	TAB 9 — VALUE
V1 MA Reversion (grid search) Fk: F1–F15; LB: 1/3/5/7/10yr; T: ±5–20% Deployed: Cu F8 5yr Al F12 5yr	dev(t) = [Fk(t) – MA_N(Fk, t)] / MA_N(Fk, t)	+1 if dev < –T (cheap) –1 if dev > +T (expensive) 0 otherwise (flat ~46–51% of days)	Lag-1 ★ (Same-Day marginally weaker)	Fixed-param WF
V2 Baz-Granger Reversal (grid search) $F1_{\text{raw}}$; LB: 1/3/5/7/10yr (cautionary alternative, not deployed)	rev(t) = $F1_{\text{raw}}(t-N) - F1_{\text{raw}}(t)$ (positive = price has fallen = cheap)	pos(t) = sign(rev(t)) +1 / –1 (no flat zone)	Same-Day ★	Fixed-param WF (collapses OOS)
TAB 10 — PORTFOLIO	TAB 10 — PORTFOLIO	TAB 10 — PORTFOLIO	TAB 10 — PORTFOLIO	TAB 10 — PORTFOLIO
Equal-Weight (EW) — Copper Mom: MA(20,45) Carry: Z-252d Value: V1-F8-5yr	port(t) = ⅓·mom_pos(t) + ⅓·carry_pos(t) + ⅓·val_pos(t)	{–1, –2/3, –1/3, 0, +1/3, +2/3, +1} Corr: Mom-Carry +0.13, Mom-Val –0.21, Carry-Val –0.04	Same-Day (all legs)	Fixed-param WF

Equal-Weight (EW) — Aluminium Mom: MA(60,115) Carry: Z-252d Value: V1-F12-5yr	$\text{port}(t) = \frac{1}{3} \cdot \text{mom_pos}(t) + \frac{1}{3} \cdot \text{carry_pos}(t) + \frac{1}{3} \cdot \text{val_pos}(t)$	{-1, -2/3, -1/3, 0, +1/3, +2/3, +1} Corr: Mom-Carry +0.23, Mom-Val -0.36, Carry-Val -0.01	Same-Day (all legs)	Fixed-param WF
Inverse-Vol (63d window) (alternative weighting, both metals)	$\sigma_i(t) = \text{rolling } 63\text{d ann. vol of leg } i\text{'s daily returns}$ $w_i(t) = (1/\sigma_i) / \sum_j (1/\sigma_j)$, $\sum w = 1$	Continuous (-1 to +1) Rebalanced daily	Same-Day (all legs)	Fixed-param WF

Performance Summary — Primary Deployed Configurations

IS = full-history backtest, net of 5bps TC unless noted. WF-OOS Sharpe (net) = average of rolling 1yr-OOS windows after a 5yr IS burn-in (IS=1260d, OOS=252d, windows labelled by end-year), the same convention used on every dashboard tab. ★ marks the configuration actually selected on the dashboard by default. All numbers re-derived directly from app.py's own functions on 2026-07-05.

Strategy / Config	Metal	IS Sharpe (gross)	IS Sharpe (net 5bps)	IS Ann Ret (gross)	Max DD	WF-OOS Sharpe (net)	OOS Methodology
Momentum MA(20,45) ★	Copper	+0.63	+0.62	+16.2%	-52%	+0.36	Fixed-param WF
Momentum MA(35,43) [legacy]	Copper	+0.72	+0.70	+18.3%	-64%	+0.47	Fixed-param WF
Momentum MA(60,115) ★	Aluminium	+0.48	+0.48	+10.7%	-50%	+0.34	Fixed-param WF
Carry Carry-Mom 20d ★ (tab default)	Copper	+0.52	+0.47	+13.5%	-47%	+0.25	Fixed-param WF
Carry Z-score 252d ★	Aluminium	+0.66	+0.64	+14.1%	-50%	+0.89	Parameter-free
Value V1 F8 5yr ±10%, Lag-1 ★	Copper	+0.33	+0.32	+4.5%	-67%	+0.77	Fixed-param WF
Value V1 F12 5yr ±10%, Lag-1 ★	Aluminium	+0.24	+0.23	+2.8%	-71%	+0.73	Fixed-param WF
EW Portfolio (3 sleeves) ★	Copper	+0.76	+0.73	+8.7%	-24%	+0.41	Fixed-param WF
EW Portfolio (3 sleeves) ★	Aluminium	+0.87	+0.85	+9.3%	-21%	+1.12	Fixed-param WF
Anchors + IS-Opt Weights (unweighted EW shown for IS context)	Copper	+0.47	+0.45	+8.7%	-46%	+0.40	Rolling WF (true OOS)

Notes on configuration choices: (1) Carry — the Copper Portfolio's Carry leg is Z-score-252d (IS net +0.23, WF-OOS net +0.07), NOT Carry-Mom-20d shown above — Carry-Mom-20d is the best STANDALONE Carry signal on the Carry tab, but it trades the CHANGE in roll yield (a momentum operation on a curve variable), which would overlap conceptually with the Momentum leg already in the blend; Z-score-252d is a standardised curve LEVEL measure, the cleaner “Carry” slot for the three-distinct-premia portfolio. Carry-Mom-20d is effectively dead on Aluminium standalone (net +0.10, WF-OOS +0.05) — Z-score-252d is both the tab default and the Portfolio leg there, so no such split exists for Aluminium. (2) Momentum — the full-history grid optimum for Copper, MA(35,43), has only an 8-day fast/slow gap (ratio 1.23) — a grid-search artifact rather than a genuine multi-timescale trend, per Prof. Ilia's trader-convention feedback. The deployed default is MA(20,45) (ratio 2.25): weaker full-sample Sharpe but a shallower drawdown and stronger walk-forward performance in the post-2022 windows. MA(35,43) remains selectable on the tab for direct comparison. (3) Value — Lag-1 (shift-2) is confirmed the production default on both metals from the dashboard's own auto-reset logic; V1's IS Sharpe is computed on this timing (Copper +0.33/+0.32 gross/net, Aluminium +0.24/+0.23) — both metals are marginally weaker under Same-Day. Inside the Portfolio (Tab 10), all three legs including Value execute Same-Day (shift-1) for uniformity across the blend — a deliberate, different convention from the Value tab's own standalone default. (4) Regime-conditional dynamic weighting (term-structure and volatility regimes, IS-optimised weights refit per rolling window) was tested walk-forward on both metals and does not reliably beat the static equal-weight blend on either metal — shelved, not deployed.

Current Positioning (latest available date, EW Portfolio)

Metal	As of	Momentum	Carry	Value	EW Portfolio
Copper	2025-12-31	+1 (long)	+1 (long)	-1 (short)	+0.33 (net long)
Aluminium	2026-05-19	+1 (long)	+1 (long)	-1 (short)	+0.33 (net long)

Data source: LME futures curve (F1–F27), LME Cash & 3M prices. Framework: Bouchoev (2024) commodity risk premia decomposition. WF-OOS avg-of-windows convention matches dashboard reporting across all tabs. Inverse-vol trailing window (63d) selected as Sharpe-optimal; EW outperforms IV on net Sharpe and max drawdown for Copper. All figures recomputed directly from app.py's own functions against current data/files.